

ZIP-response screening with the ZIP simulation design

mfscreen simulation

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1 Objective

This report reproduces the simulation framework of `3_ZIP_abs_Simulations_screening_Version` pour `Vincent.Rmd`. The response is the original zero-inflated Poisson response

$$Y = Y_1 Y_2,$$

where Y_1 is Bernoulli and Y_2 is Poisson with a conditional intensity defined below.

The output follows the same structure as the ZIP report: selection frequencies for $X_1, \dots, X_{18}$, the true-positive rate, the false-positive rate, and the mean cardinality of the selected set. We also add FDR-SSP curves comparing `mfscreen` with three marginal screening competitors.

2 Mathematical simulation framework

Let $n = 1500$, $p = 2000$, $D = 10$, and $N = 80$ Monte Carlo replications. For observation i , write

$$X_i = (X_i^{(1)}, \dots, X_i^{(p)}).$$

The active response-generating variables are $X^{(1)}, \dots, X^{(10)}$. As in the ZIP screening table, the variables counted as positives are

$$\mathcal{A} = \{1, \dots, 14\},$$

because $X^{(11)}, \dots, X^{(14)}$ are noisy copies of active Gaussian variables. The null set used for false-positive and FDR calculations is

$$\mathcal{A}^c = \{15, \dots, 2000\}.$$

2.1 First five Gaussian variables

Let

$$(X^{(1)}, \dots, X^{(5)}) \sim \mathcal{N}_5(0, \Sigma_\rho), \quad (\Sigma_\rho)_{jk} = \rho^{|j-k|}.$$

The executable parameter setting in the source ZIP file is

$$\rho = 0.$$

2.2 Additional variables entering the response

Independently of the first five Gaussian variables, generate

$$X^{(6)} \sim \text{Bernoulli}(0.35), \quad X^{(7)} \sim \chi_2^2,$$

$$X^{(8)} = X^{(1)}P, \quad P \sim \text{Poisson}(2),$$

$$X^{(9)} = X^{(2)}G, \quad G \sim \mathcal{N}(1, 1),$$

and

$$X^{(10)} \sim t_{14}.$$

2.3 Correlated non-generating variables

Let

$$Z_3 \sim \mathcal{N}(0, 0.4^2), \quad Z_4 \sim \mathcal{N}(0, 0.02^2), \quad Z_5 \sim \mathcal{N}(0, 0.35^2), \quad Z_6 \sim \mathcal{N}(0, 0.55^2),$$

independent of the variables above. Define

$$X^{(11)} = X^{(1)} + Z_3, \quad X^{(12)} = X^{(3)} + Z_4, \quad X^{(13)} = X^{(4)} + Z_5, \quad X^{(14)} = X^{(5)} + Z_6,$$

and

$$X^{(15)} = Z_3, \quad X^{(16)} = Z_4, \quad X^{(17)} = Z_5, \quad X^{(18)} = Z_6.$$

The variables $X^{(11)}, \dots, X^{(14)}$ are correlated with active variables but do not enter the response equation. Variables $X^{(15)}, \dots, X^{(18)}$ are null variables and are counted as false positives when selected.

2.4 Remaining null block

The remaining variables

$$X^{(19)}, \dots, X^{(2000)}$$

are independent of $(Y, X^{(1)}, \dots, X^{(18)})$. They form a stationary Gaussian sequence with covariance

$$\text{Cov}(X^{(18+j)}, X^{(18+k)}) = \rho^{|j-k|}, \quad \rho = 0.7.$$

2.5 Zero-inflated Poisson response

The coefficient vector is the one used in the executable part of the ZIP file:

$$\theta^* = (1, 2, \dots, 11)^T.$$

For the first ten covariates, define the linear predictor

$$\eta_i = \theta_0^* + \sum_{j=1}^{10} \theta_j^* X_i^{(j)}.$$

The zero-inflation component is

$$Y_{1i} \mid X_i \sim \text{Bernoulli}(\pi_i), \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}.$$

The Poisson component is

$$Y_{2i} \mid X_i \sim \text{Poisson}(\lambda_i), \quad \lambda_i = |\eta_i|.$$

Conditionally on X_i , Y_{1i} and Y_{2i} are generated independently. The observed response is

$$Y_i = Y_{1i} Y_{2i}.$$

Thus

$$\mathbb{P}(Y_i = 0 \mid X_i) = (1 - \pi_i) + \pi_i \exp(-|\eta_i|),$$

and for every integer $y \geq 1$,

$$\mathbb{P}(Y_i = y \mid X_i) = \pi_i \exp(-|\eta_i|) \frac{|\eta_i|^y}{y!}.$$

3 Screening rules and criteria

For **mfscreen**, the package returns reciprocal scores D_j^2 , with smaller values indicating stronger marginal association. At nominal level q , variable j is selected when

$$D_j^2 \leq \frac{1}{\gamma_q^2}, \quad \gamma_q = \Phi^{-1}(1 - q/2).$$

The ZIP-type screening table uses the source-file threshold

$$q_n = 0.15.$$

For a selected set $\widehat{\mathcal{M}}$, define

$$\text{TPR} = \frac{1}{|\mathcal{A}|} \sum_{j \in \mathcal{A}} \mathbf{1}\{j \in \widehat{\mathcal{M}}\},$$

$$\text{FPR} = \frac{1}{|\mathcal{A}^c|} \sum_{j \in \mathcal{A}^c} \mathbf{1}\{j \in \widehat{\mathcal{M}}\},$$

$$\text{FDR} = \frac{\sum_{j \in \mathcal{A}^c} \mathbf{1}\{j \in \widehat{\mathcal{M}}\}}{\max(1, |\widehat{\mathcal{M}}|)},$$

and

$$\text{SSP} = \mathbf{1}\{\mathcal{A} \subseteq \widehat{\mathcal{M}}\}.$$

The three competitors are:

- Pearson SIS, based on marginal Pearson correlations with the observed ZIP response;
- Spearman SIS, based on marginal rank correlations with the observed ZIP response;
- Poisson-score SIS, based on the univariate Poisson score statistic for a count-response marginal model.

We also add Poisson-Wald SIS. It uses the same univariate Poisson marginal model as Poisson-score SIS, but fits the slope and uses a Wald statistic.

For the FDR-SSP curves, all methods, including `mfscreen`, select the top k variables according to their marginal ranking score. This makes the curves directly comparable at the same selected model sizes. The calibrated `mfscreen` threshold $q_n = 0.15$ is reported separately in the ZIP-type screening table.

3.1 Mathematical formulation of the competitors

For each covariate j , write

$$X_j = (X_1^{(j)}, \dots, X_n^{(j)})^T, \quad Y = (Y_1, \dots, Y_n)^T.$$

Pearson SIS. The marginal Pearson statistic is

$$R_j = \frac{\sum_{i=1}^n (X_i^{(j)} - \bar{X}_j)(Y_i - \bar{Y})}{\left[\sum_{i=1}^n (X_i^{(j)} - \bar{X}_j)^2\right]^{1/2} \left[\sum_{i=1}^n (Y_i - \bar{Y})^2\right]^{1/2}}.$$

Variables are ranked by $|R_j|$.

Spearman SIS. Let $r_i^{(j)}$ be the rank of $X_i^{(j)}$ among $X_1^{(j)}, \dots, X_n^{(j)}$, and let s_i be the rank of Y_i among $Y_1, \dots, Y_n$. The Spearman statistic is

$$R_j^S = \frac{\sum_{i=1}^n (r_i^{(j)} - \bar{r}_j)(s_i - \bar{s})}{\left[\sum_{i=1}^n (r_i^{(j)} - \bar{r}_j)^2\right]^{1/2} \left[\sum_{i=1}^n (s_i - \bar{s})^2\right]^{1/2}}.$$

Variables are ranked by $|R_j^S|$.

Poisson-score SIS. For each j , consider the marginal Poisson GLM

$$Y_i \mid X_i^{(j)} \sim \text{Poisson}(\mu_{ij}), \quad \log \mu_{ij} = \alpha_j + \beta_j X_i^{(j)}.$$

The null hypothesis is $H_{0j} : \beta_j = 0$. Under the null model,

$$\hat{\alpha}_{0j} = \log \bar{Y}, \quad \hat{\mu}_{0i} = \bar{Y}.$$

Using centered covariates, the score for β_j at H_{0j} is

$$U_j = \sum_{i=1}^n (X_i^{(j)} - \bar{X}_j)(Y_i - \bar{Y}),$$

and the Fisher information for β_j at the null is

$$I_j = \bar{Y} \sum_{i=1}^n (X_i^{(j)} - \bar{X}_j)^2.$$

The standardized score statistic is

$$Z_j^{\text{score}} = \frac{U_j}{\sqrt{I_j}},$$

with two-sided p-value

$$p_j^{\text{score}} = 2\Phi(-|Z_j^{\text{score}}|).$$

Poisson-Wald SIS. In the same marginal Poisson GLM, fit the unrestricted model and obtain the maximum likelihood estimate $(\hat{\alpha}_j, \hat{\beta}_j)$. Let $\widehat{\text{se}}(\hat{\beta}_j)$ be the standard error of $\hat{\beta}_j$, computed from the inverse observed Fisher information. The Wald statistic is

$$Z_j^{\text{Wald}} = \frac{\hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)},$$

with two-sided p-value

$$p_j^{\text{Wald}} = 2\Phi(-|Z_j^{\text{Wald}}|).$$

The Poisson-score and Poisson-Wald tests use the same marginal Poisson model and test the same null hypothesis $H_{0j} : \beta_j = 0$. They have the same standard normal limiting null distribution. They are not exactly the same in finite samples: the score statistic is evaluated at the restricted null model, whereas the Wald statistic is evaluated at the fitted unrestricted model.

4 Monte Carlo computation

For each replication, one full design matrix and one ZIP response are generated. The same data set is used for `mfscreen` and all competitors.

5 ZIP-type screening output

The following table has the same structure as the output table in the ZIP simulation file. It reports the selection frequency for $X_1, \dots, X_{18}$ at $q_n = 0.15$, followed by the average true-positive rate, the average false-positive rate, and the average size of the selected set.

Table 1: ZIP-design screening results: $N = 80$, $n = 1500$, $q_n = 0.15$.

quantity	value
X1	1.000
X2	1.000
X3	1.000
X4	1.000
X5	1.000
X6_bern	0.950
X7_chisq2	1.000
X8=X1*P(2)	1.000
X9=N(1,1)*X2	1.000
X10=Student(14)	1.000

quantity	value
X11=X1+Z3	1.000
X12=X3+Z4	1.000
X13=X4+Z5	1.000
X14=X5+Z6	1.000
X15=Z3	0.162
X16=Z4	0.150
X17=Z5	0.138
X18=Z6	0.212
taux_vrais_positifs	0.996
taux_faux_positifs	0.152
cardinal_moyen_ensemble_indices_selectionnees	316.575

The expected cardinality under perfect recovery of the 14 positive variables and nominal false-positive rate $q_n = 0.15$ is

$$14 + 0.15 \times 1986 = 311.9.$$

Table 2: Global mfscreen metrics at the source threshold $q_n = 0.15$.

metric	value
FDR	0.956
FPR	0.152
TPR	0.996
SSP	0.950
Mean selected cardinality	316.575

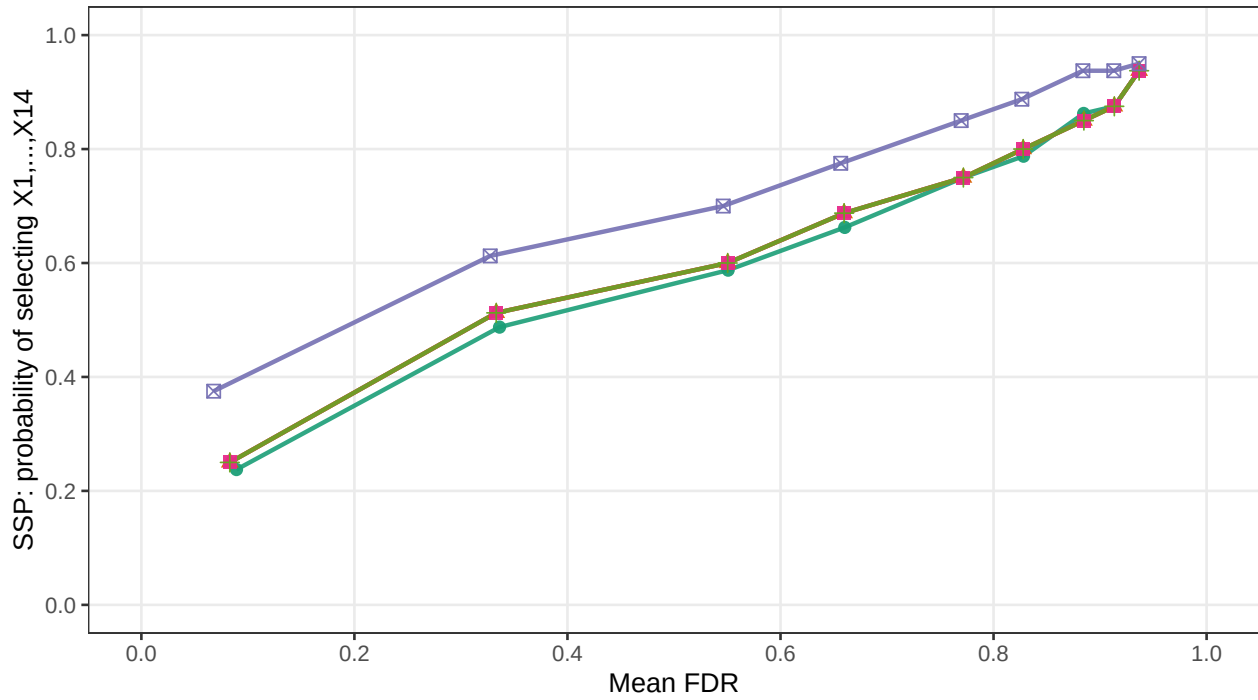
6 Competitor comparison

Table 3: FDR-SSP curve summary over $N = 80$ replications.

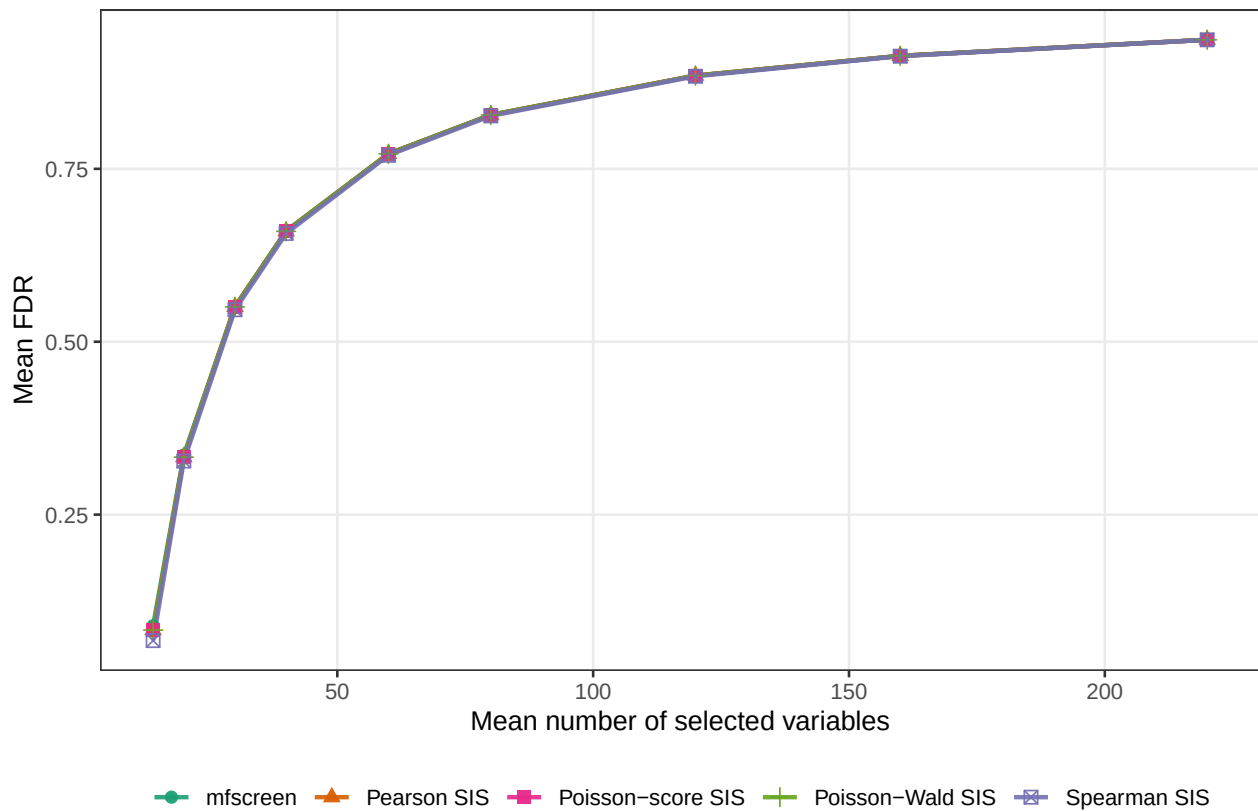
	method	parameter_label	fdr	ssp	tpr	selected_count
2	mfscreen	top 14	0.089	0.238	0.911	14
4	mfscreen	top 20	0.336	0.488	0.948	20
6	mfscreen	top 30	0.551	0.588	0.963	30
7	mfscreen	top 40	0.660	0.662	0.971	40
8	mfscreen	top 60	0.772	0.750	0.978	60
9	mfscreen	top 80	0.828	0.787	0.982	80
1	mfscreen	top 120	0.885	0.863	0.989	120
3	mfscreen	top 160	0.913	0.875	0.990	160
5	mfscreen	top 220	0.937	0.938	0.996	220
11	Pearson SIS	top 14	0.083	0.250	0.917	14
13	Pearson SIS	top 20	0.333	0.512	0.953	20
15	Pearson SIS	top 30	0.550	0.600	0.963	30
16	Pearson SIS	top 40	0.660	0.688	0.972	40
17	Pearson SIS	top 60	0.772	0.750	0.979	60
18	Pearson SIS	top 80	0.828	0.800	0.984	80
10	Pearson SIS	top 120	0.885	0.850	0.988	120

	method	parameter_label	fdr	ssp	tpr	selected_count
12	Pearson SIS	top 160	0.913	0.875	0.990	160
14	Pearson SIS	top 220	0.937	0.938	0.996	220
20	Poisson-score SIS	top 14	0.083	0.250	0.917	14
22	Poisson-score SIS	top 20	0.333	0.512	0.953	20
24	Poisson-score SIS	top 30	0.550	0.600	0.963	30
25	Poisson-score SIS	top 40	0.660	0.688	0.972	40
26	Poisson-score SIS	top 60	0.772	0.750	0.979	60
27	Poisson-score SIS	top 80	0.828	0.800	0.984	80
19	Poisson-score SIS	top 120	0.885	0.850	0.988	120
21	Poisson-score SIS	top 160	0.913	0.875	0.990	160
23	Poisson-score SIS	top 220	0.937	0.938	0.996	220
29	Poisson-Wald SIS	top 14	0.083	0.250	0.917	14
31	Poisson-Wald SIS	top 20	0.333	0.512	0.953	20
33	Poisson-Wald SIS	top 30	0.550	0.600	0.963	30
34	Poisson-Wald SIS	top 40	0.660	0.688	0.972	40
35	Poisson-Wald SIS	top 60	0.772	0.750	0.979	60
36	Poisson-Wald SIS	top 80	0.828	0.800	0.984	80
28	Poisson-Wald SIS	top 120	0.885	0.850	0.988	120
30	Poisson-Wald SIS	top 160	0.913	0.875	0.990	160
32	Poisson-Wald SIS	top 220	0.937	0.938	0.996	220
38	Spearman SIS	top 14	0.068	0.375	0.932	14
40	Spearman SIS	top 20	0.328	0.613	0.961	20
42	Spearman SIS	top 30	0.546	0.700	0.972	30
43	Spearman SIS	top 40	0.657	0.775	0.981	40
44	Spearman SIS	top 60	0.770	0.850	0.988	60
45	Spearman SIS	top 80	0.827	0.887	0.991	80
37	Spearman SIS	top 120	0.884	0.938	0.996	120
39	Spearman SIS	top 160	0.913	0.938	0.996	160
41	Spearman SIS	top 220	0.937	0.950	0.996	220

7 FDR-SSP curves



Each method varies the same top-k selected model size grid.



8 Conclusion

The ZIP-response experiment uses the ZIP-design covariates without changing the active/null structure of the screening table. At the source threshold $q_n = 0.15$, the mfscreen false-positive rate is directly comparable with the nominal target, while the reported selection frequencies show which of $X_1, \dots, X_{18}$ are retained. The FDR-SSP curves compare all methods at the same top- k selected model sizes, so the curves are directly comparable.